Technical note

Detecting forecast busts in a medium-range ensemble

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This note describes the method and evaluation protocol behind a request for access to NEPS output. It is written for a reader familiar with ensemble verification.

1. Motivation

Operational practice treats ensemble spread as a proxy for forecast confidence. We measured how well that proxy performs over India, against IMD 0.25° gauge-gridded daily rainfall colocated onto the forecast grid by exact float match — no interpolation.

Pooled across all forecasts, the log spread–log error correlation accounts for roughly 44% of the variance in forecast error. Conditioned on rainfall magnitude — comparing forecasts that predict similar amounts — it accounts for between 0.17% and 0.90%, depending on how a "wet" gridpoint is defined. The pooled figure largely reflects that both spread and error grow with rainfall amount. It is a property of the rainfall distribution, not evidence that spread identifies which forecasts will fail.

That gap is the opening this work addresses: the quantity forecasters rely on carries far less information about failure than its pooled statistics suggest.

2. Data and labelling

- Predictors: NOAA GEFSv12 reforecast (Guan et al., 2022, Mon. Wea. Rev. 150(3), 647–665), 11 members, 0.5°, Day 1 to Day 10.
- Truth: IMD 0.25° gauge-gridded daily rainfall. A gauge analysis, not a reanalysis, so no model physics is shared with the forecast being judged.
- Domain: 1,252 gauge-observed land gridpoints of the 5,041-point analysis box. The gauge mask is stated rather than filled in.
- Bust definition: |ensemble mean − observed| above the 95th percentile, fitted per region and lead day ON THE TRAINING WINDOW ONLY, then applied unchanged. Base rate is about 6%.
- Study window: 2015–2019, 256 usable initialisations, 3,205,120 labelled gridpoint-days.

3. Model

Twenty-three features, every one derived from the ensemble forecast and none from the truth: member means and spreads of 500 hPa geopotential height and winds, 850 hPa winds, precipitable water, CAPE and precipitation; derived quantities (500–850 hPa shear, flow deformation, height gradient); and anomalies against a climatology fitted on the training window only. Latitude, longitude and region are deliberately excluded so the model cannot memorise where busts happen.

Gradient-boosted trees, CPU only, no tuning. Probabilities are calibrated by isotonic regression fitted on a held-out validation slice, never on test data.

4. Evaluation protocol

Two choices in the protocol matter more than the model.

Stratification. ROC-AUC is computed within rainfall-magnitude bins and recombined n-weighted. Binning is on forecast rainfall — known at forecast time — never on the observation. Pooled AUC rewards any score that tracks rainfall amount, which flatters the incumbent for a reason unrelated to skill at detecting failure.

Cluster bootstrap. Intervals resample whole initialisation dates, not gridpoints. Gridpoints within one forecast are nearly the same observation; a row-level bootstrap would report a falsely tight interval.

Accuracy is deliberately not reported anywhere. At a 6% base rate, predicting "never a bust" scores 94% and is useless.

5. Results

Configuration	Stratified ROC-AUC	Notes
Ensemble spread (incumbent)	**0.565**	52 test dates, 2019, held out entirely
VISHWAS	**0.728**	same dates, same truth, same bins
Margin	**+0.13 to +0.16**	95% CI +0.108 to +0.193; ahead in all 8 bins
Five-year replication	**+0.149**	104 dates, 2018–19; 95% CI +0.116 to +0.203
Difference between the two	**−0.012**	paired; 95% CI −0.051 to +0.011, spans zero

The model is ahead of ensemble spread at every lead day from 1 to 10, with the whole confidence interval above zero at each; the margin eases from +0.171 at Day 1 to +0.149 at Day 10. At a matched alert budget — the same number of flagged gridpoints — it recovers 64% of busts against ensemble spread's 55%.

The five-year row is the honest headline. Tripling the training data left the result statistically indistinguishable from the original: one year of training was already sufficient, and that is now a measured result rather than an assumption.

6. Operational feasibility

The published model scores a live operational GEFS cycle without modification: 1,210 byte-range reads, 208 MB, 90 seconds end to end on a laptop, with the 31-member operational ensemble subsampled to the 11 members the model was trained on. Features are built by the same code path as the reforecast, and the stored climatology generalises to any date without refitting. No GPU, no paid API, no credentials.

7. Limitations, stated plainly

- The ensemble is GEFS, not NEPS. NEPS reforecast archives are not publicly accessible; GEFS is the only medium-range ensemble with a public multi-decade reforecast. This is a substitution, not a detail, and it is the reason for the accompanying request.
- The domain is the IMD gauge grid — 1,252 of 5,041 gridpoints. Beyond it, no label has ever existed, so nothing there has been or can be verified.
- The reforecast carries 11 members where the operational system carries 31. A noisier spread estimate is easier to beat, and we report that rather than leave it to be found.
- This is detection, not correction. The system flags forecasts likely to fail. It does not adjust them, and no claim is made that it could.
- Verification is against rainfall only. Other variables are untested.

8. What NEPS access would enable

The method consumes ensemble mean and spread and is agnostic to the model that produced them. With NEPS output it could be retrained and re-verified on the ensemble NCMRWF operates, which would establish whether the result transfers to India's own system — the question we cannot presently answer, and the one that matters most for any operational use. Required fields and volumes are listed in Annexure A of the covering letter.

Prototype dashboard: vishwas-five.vercel.app · Code and full verification records available to NCMRWF on request.